

ADDENDUM TI MANUAL TTLS PACKING INSTRUCTIONS



August 1, 2019

#020-00077 REV C

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ENGINEERING CHANGE RECORD**ADDENDUM TI MANUAL TRANSIT CASE PACKING INSTRUCTIONS****DOCUMENT #020-00077 REV C**

Revision	Issue Date	Revision Description	ECN #
A	2015Jul07	Initial Release	1940
B	2017Jul18	More details added	02046
C	2019Aug1	More details added	ECA-00148

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PACKING THE TRANSIT CASES

1. Packing list found in Appendix A contains an itemized list of contents and part numbers for each transit case. Print and laminate two copies of packing list. One copy will be used to pack a page with each corresponding transit case. The second copy will be right aligned and made into a book with a comb and book binder.
2. Complete container packing checklist found in Appendix B.
3. Move three 890-00105, three 890-00104, four 890-00120, and four 890-00107 to suite 101.
4. Print and place box labels. Use label template found in PDXeprt under PN 180-00015. Identity box contents on the right corner of lid and box number on left corner. Repeat labeling on the profile side of each case. There should now be four labels on each box, 56 labels total.
5. When packing, add foam to pad contents. Remember these boxes are traveling across the world, pack accordingly.
6. Remove unnecessary plastic and other FOD from all ECS cases and other items packed directly into the container.
7. Record the serial numbers of major items throughout packing procedure.
8. Remove identifying stickers that trace transit cases or items to manufactures and vendors.

1.1 Spare Sensor

Box: 1

Transit Case: 890-00105

Contents: Spare sensor

1. Wrap sensor in ESD safe bag and secure contents with protective foam.
2. Record serial number.



Figure 1 Spare Sensor in ESD Safe Bag

1.2 RCUs and Monitors

Box: 2, 3

Transit Case: 890-00104

Contents: Monitor, monitor stand, mouse and keyboard, RCU, RCU bag, VGA display and power, and RCU Ethernet hub (only for box 3)

1. Place two monitor stands in box #2 with the bases on the east and west walls.
2. Place two monitors in the north and south walls of the case, with screens facing out.
3. Place two RCU computers in case between the monitors, separated from the monitors with foam strips.
4. Place remaining RCU items in case, around the RCU computers.
5. Add more protective foam if necessary.
6. Repeat steps 1-5 for box #3, and include the Ethernet hub.



Figure 2 RCUs and Monitors



Figure 3 RCUs and Monitors with Ethernet Hub

1.3 Radios

Box: 4

Transit Case: 890-00120

Contents: RF and other relevant cables, ground to air radio hardware, pair wireless ACS, directional antennas, wireless radio attachment and install hardware, wireless power bricks, and COM radio antenna cable

1. Secure and pad contents with protective foam.



Figure 4 Radios

1.4 FTM and Total Station

Box: 5

Transit Case: 890-00105

Contents: Theodolite laser, safety vest, FM transceiver, fiberglass tape, tripod lights, FTM cable, batteries

1. Secure and pad contents with protective foam.
2. Attach the end of FTM cable to a secure location.



Figure 5 FTM and Total Station

1.5 Cal/BIT and LNA

Box: 6

Transit Case: 890-00120

Contents: Cal/BIT, LNA (Low Noise Amplifier), tilt sensors, ob-light cable, LNA and Cal/BIT cable

1. Place one set of the cables in both the LNA and the Cal/BIT
2. Secure and pad contents with protective foam.
3. Confirm there are no points of contact between items. This will prevent scratching during transit.
4. Record serial numbers for all items.



Figure 6 Cal/BIT, LNA, and Tilt Sensor Assembly

1.6 Sand Coasters

Box: 7

Transit Case: 890-00120

Contents: Sand coasters, rope, and plates

1. Place heavier items (plates) in the bottom on the case.
2. Secure and pad contents with protective foam.



Figure 7 Sand Coasters and Rope

1.7 Tripod Base

Box: 8

Transit Case: 890-00104

Contents: Tripod bases

1. Secure and pad contents with protective foam.
2. If necessary, make cuts in the case's foam to allow all three bases to fit.



Figure 8 Tripods Bases

1.8 Spare Cal/BIT

Box: 9

Transit Case: 890-00120

Contents: Spare Cal/BIT and Portable Navigation Analyzer

1. Place bagged Cal/BIT Spare in case. Secure and pad with protective foam.
2. Pad the Portable Navigation Analyzer with a liberal amount of foam.
3. Record serial numbers of all items.



Figure 9 Spare Cal/BIT and Portable Navigation Analyzer

1.9 Tool Kits

Box: 10

Transit Case: 890-00105

Contents: Makita impact driver, Makita drill/driver set, hand tools kit, rain hood, fluke clamp meter, tool kits, consumables kit

1. For complete list of tools and consumables, see the Packing List document
2. Protective foam may not be necessary for this case.



Figure 10 Tool Kits

1.10 Theodolite Tripod

Box: 11

Transit Case: 890-00107

Contents: ADIRpro prism, tension gauge, prism pole, Rami ground to air antenna and rods, tripod, mount rods, carabiner and rope assemblies

1. Secure and pad contents with protective foam.



Figure 11 Theodolite Tripod and Prism Equipment

1.11 Anchoring

Box: 12, 13, 14

Transit Case: 890-00107

Contents of 12: ESA guywire bracket, ASA turnbuckles, antenna structure turnbuckles, augers, and arrowhead anchors

Contents of 13: Winch assembly, mobile ground sticks kit, poles, arrowhead hammer, and air to ground antenna mount

Contents of 14: Gin pole auger, turnbuckles, shackles, 2' chains, shackle and cable assemblies, spar bracket, assorted stakes, and loops

1. Protective foam is not necessary for boxes #12, #13, and #14.
2. Pack contents to distribute the weight evenly.



Figure 12 Anchoring Case (Box 12)



Figure 13 Anchoring Case (Box 13)



Figure 14 Anchoring Case (Box 14)

PACKING THE CONTAINER

2.1 Pre-Packing Procedure

1. Mount antennas according to wall labels. For questions, reference drawing 920-00713.
2. Tighten cover screws on the back of the racks. See Figure 15.
3. Mount tripod brackets on the wall. Confirm bracket is threaded halfway along rod. See Figure 17 and drawing 920-00767 for reference.
4. Strap AC units down.
5. Confirm rack cables are routed in an organized fashion, similar to Figure 16.
6. Turn off the air conditioners and tape remotes in place.
7. Power off all UPS.
8. Power off all other electronics.



Figure 15 Cover Screw Instructions

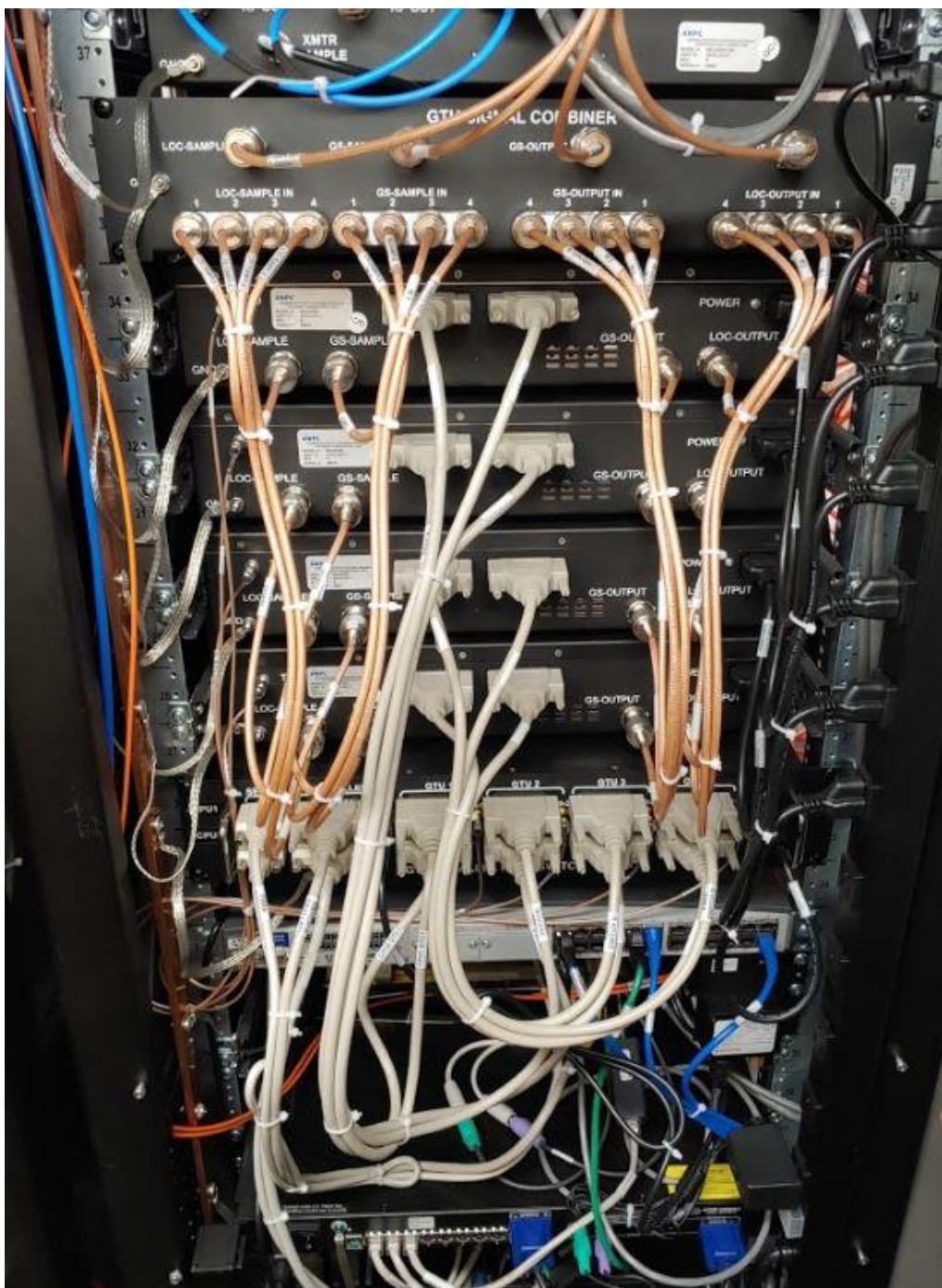


Figure 16 Organized Rack Cables



Figure 17a Mounting Tripod Brackets



Figure 17b



Figure 17c

2.2 Behind the Racks

1. Start packing the back of the shelter first. Transit cases can be stacked behind the racks. Boxes 1-8 will form the first stacks, as shown in the table below. This view is in the y-z plane, facing the back wall from racks. See Figure 18, Figure 19, and Figure 20 for more details.

	BOX 3	BOX 6
BOX 4	BOX 2	BOX 1
BOX 7	BOX 8	BOX 5



Figure 18 First Column of Boxes



Figure 19 Second Column of Boxes



Figure 20 Ratchet Straps Securing Boxes

2. Secure boxes 1-8 with ratchets straps, as shown in Figure 20 and Figure 22.
3. Pad the sides of box 4 and 7 against the racks and antennas.
4. If necessary, add an attachment point to secure box 4 and 7. Drill a hook, as shown in Figure 21, to the floor in front of the transformer.



Figure 21 Hook for Boxes #4 and #7



Figure 22 Boxes #4 and #7

2.3 The Tower Sections

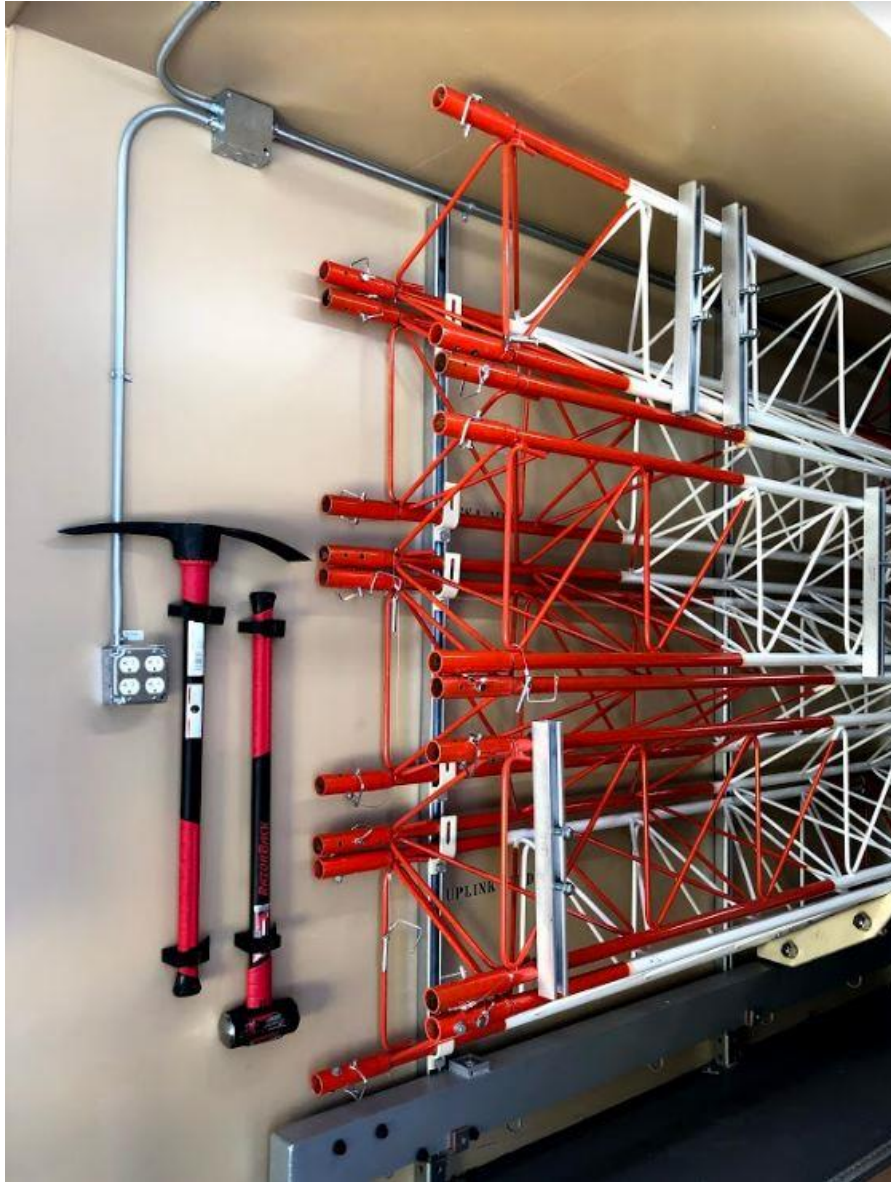


Figure 23 Mounted Tower Sections

5. Secure tower sections to the shelter wall. The middle section of wall mounts may need to be removed to accommodate the tower section orientations. Load sections so the bases are on one side, and tops on the other. See Figure 23.
6. Mount pick and sledgehammer to wall as shown in Figure 23.
7. Slide poles through tower sections and secure with straps. See Figure 24.
8. Secure climbing harness to tower sections as shown in Figure 25.

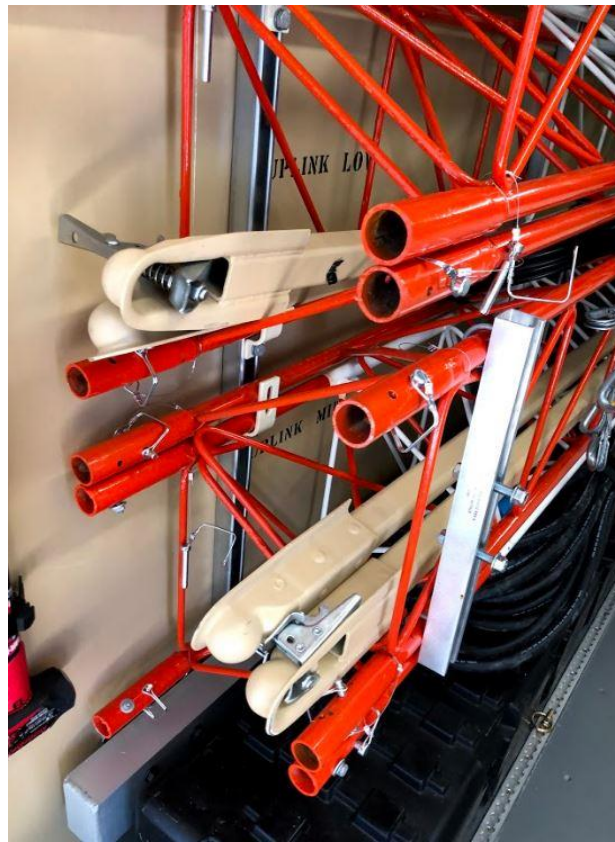


Figure 24 Loaded Poles



Figure 25 Climbing Harness

9. Secure cables and guywires to towers. Use 140-00176 and 140-00177 straps and confirm nothing is loose. See Figure 26 and Figure 27.



Figure 26 Mounted Cables



Figure 27 Fastening Cables

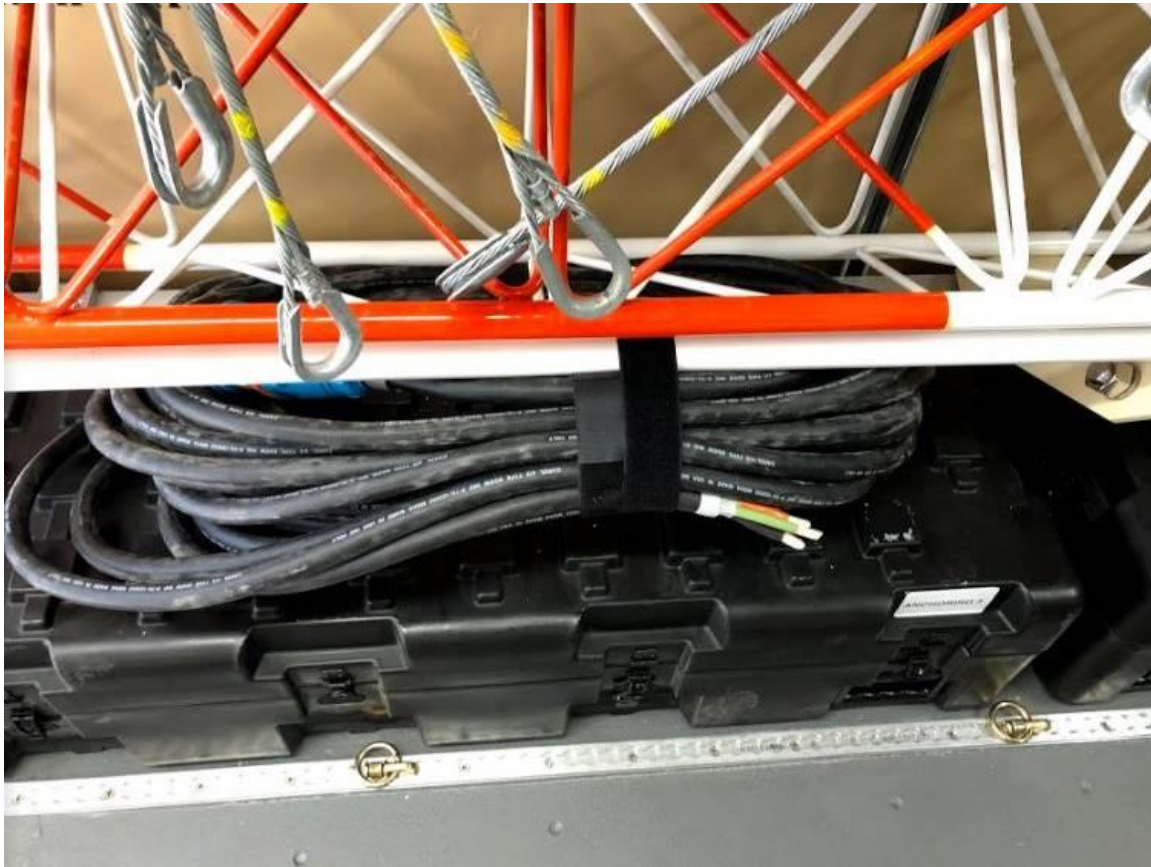


Figure 28 Boxes 12 and 14 Under the Towers

10. Pack and secure boxes 12 and 14 under the tower sections. See Figure 28.

2.3 ASA and ESA Cabinets

11. Set up yellow ramps on the edge of shelter entrance. Load ESA and ASA cabinets into the container. Orientate shelters according to Figure 29 and bolt the c-channel to the ground. See Figure 30.
12. Rotate all eight brackets on cabinets so transit lids will not rest on the brackets.



Figure 29 ASA and ESA Shelter Orientation with Transport Lids



Figure 30 ASA and ESA Shelters Bolted to the Ground

13. Load boxes 11 and 13 onto shelter lids and secure with ratchet straps to floor. See Figure 31.
14. Slide trolley under the ASA cabinet, as shown in Figure 32.



Figure 31 Box #12 Loaded onto Shelters



Figure 32 Trolley Under ASA Cabinet

2.4 The Doors

15. Roll remaining cables onto reels, taking care to roll the cable tightly onto the reels to make all of the cable fit. The 100-meter cables will be packed onto two separate reels. In the case of 100-meter cables, start rolling from the ends as shown in Figure 34.
16. When rolling the three shelter cables, attach them to the spools as shown in Figure 35, Figure 36, and Figure 37.
17. Only pull the cables by gripping the metal strain-relief.
18. NEVER PULL THE CABLE USING THE FIBER OPTIC CONNECTOR.



Figure 33 100-meter Cable



Figure 34



Figure 35



Figure 36



Figure 37

19. Load spooled cables and ESA and uplink bases onto the shelter doors, as shown in Figure 38 and Figure 39.
20. Mount a shovel to each door, as shown in Figure 38 and Figure 39.

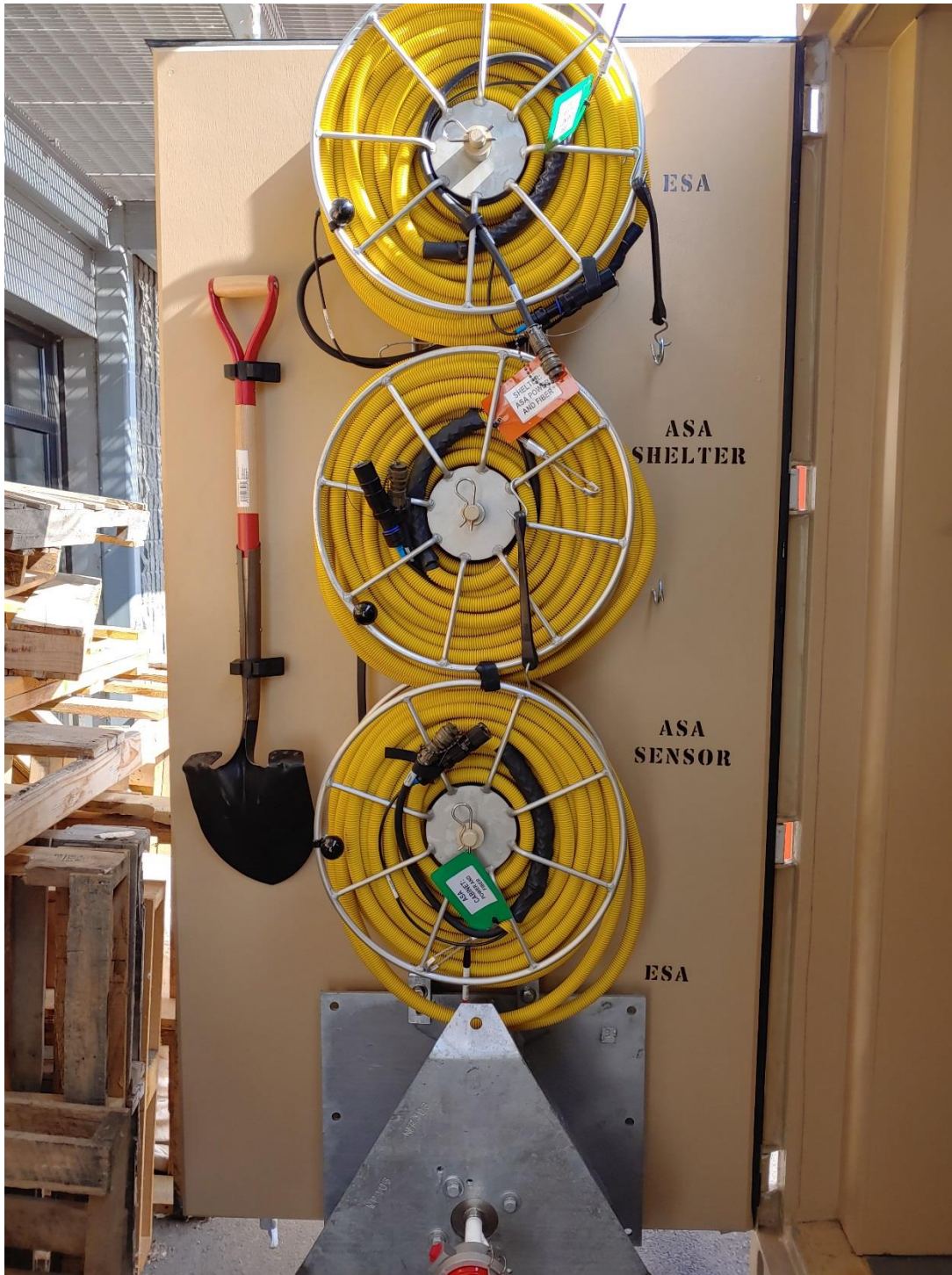


Figure 38 Loaded Cables and ESA Base



Figure 39 Loaded ATA Cable, Uplink Base, and Shovel

2.5 The Entrance

21. Load Cal/BIT spool and orientate according to Figure 40. The generator is pictured for reference.
22. Secure Cal/BIT spool as shown in Figure 41, Figure 42, and Figure 43.



Figure 40 Cal/BIT Spool



Figure 41



Figure 42



Figure 43

23. Load the generator in front of the ASA cabinet. Bolt to the ground as shown in Figure 44.



Figure 44 Generator Bolted to the Ground

24. Place transit lid on top of the generator and load boxes 9 and 10 onto the lid. See Figure 45.



Figure 45 Generator Loaded with Boxes #9 and #10

25. Load fuel drum between boxes 9, 11, and 13. See Figure 46.

26. Secure fuel drum according to Figure 47.



Figure 46 Loaded Fuel Drum

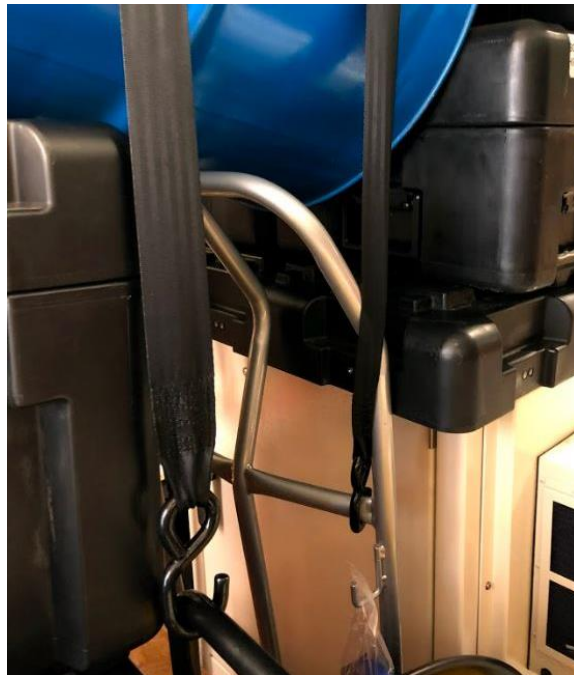


Figure 47 Secured Fuel Drum

27. Load large sand coasters to the right of generator. See Figure 48.
28. Load small sand coasters under generator. See Figure 49.
29. Secure sand coasters with ratchet straps.
30. Place yellow ramps in container.



Figure 48 Secured Large Sand Coasters



Figure 49 Secured Small Sand Coasters.



Figure 50 Entrance of Loaded Container

LOADING ONTO A C-17

Follow the load master commands to drive the F-650 truck onto the C-17. Dump the air bags. Maneuver to the center part of the aircraft under the wing. Secure with chains.



Figure 51

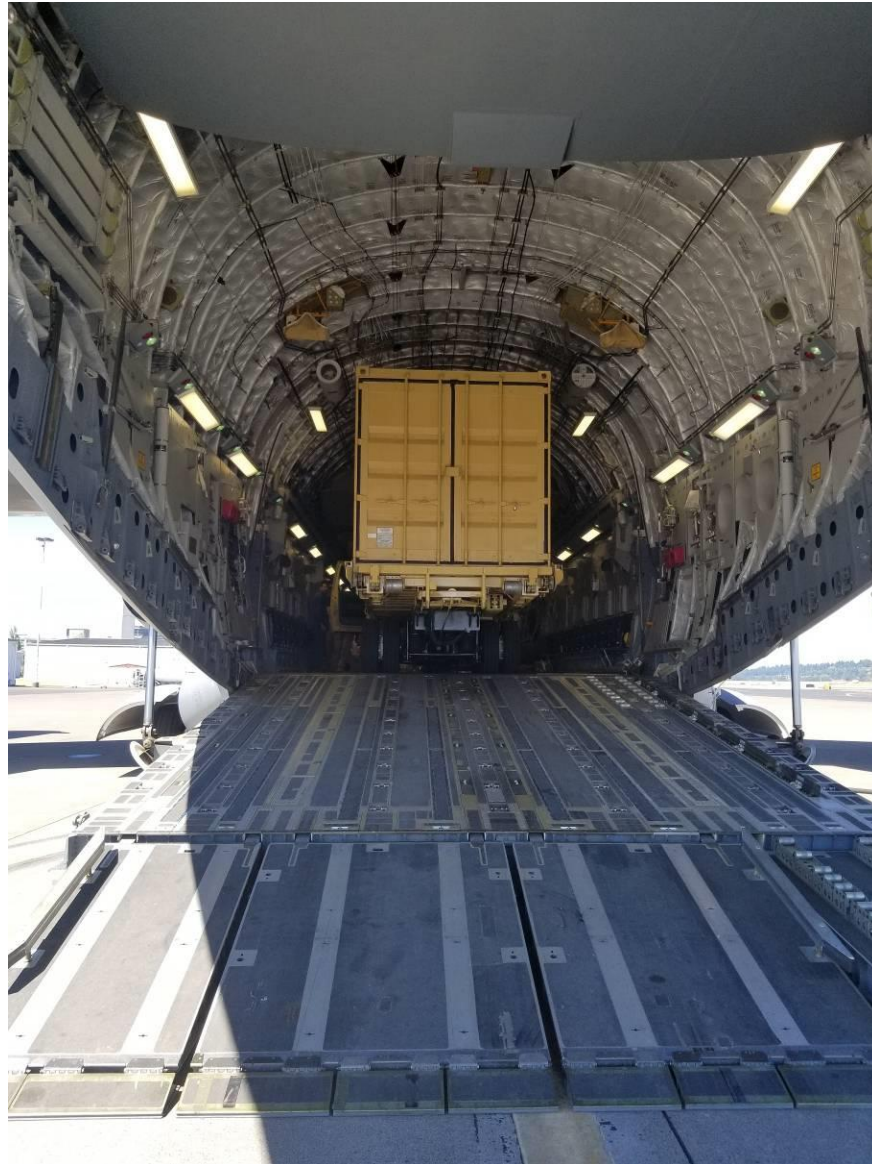


Figure 52



Figure 53 Loading Complete with Vehicle Chained Securely

APPENDIX A

Box 1	Spare sensor
Box 2	RCU's and Monitors
Box 3	RCU's and Monitors
Box 4	Radios
Box 5	FTM and Total Station
Box 6	Cal/Bit LNA
Box 7	Sand coasters
Box 8	Tripod Base
Box 9	Spare Cal/Bit
Box 10	Tool Kits
Box 11	Theo Tripod
Box 12	Anchoring 1
Box 13	Anchoring 2
Box 14	Anchoring 3

Packing List				
Tool Kit		Room		
Part #	Description	Bag #1	Bag #2	Total
730-00245	SKT DEEP 3/8" SQ DR 6-POINT 1/2"	2	2	4
730-00246	SKT DEEP 3/8" SQ DR 6-POINT 9/16"	2	2	4
730-00247	SKT DEEP 3/8" SQ DR 6-POINT 5/8"	2	2	4
730-00248	SKT DEEP 3/8" SQ DR 6-POINT 11/16"	2	2	4
730-00249	SKT DEEP 3/8" SQ DR 6-POINT 3/4"	2	2	4
730-00250	SKT DEEP 3/8" SQ DR 6-POINT 15/16"	2	2	4
730-00251	WRENCH RATCHET 3/8" SQ DRIVE 10" L	2	2	4
730-00252	WRENCH 9/16" COMBO	2	2	4
730-00253	WRENCH 13/16" COMBO	2	2	4
730-00254	SCDR 1/4" FLAT 8" L	1	1	2
730-00255	SCDR #2 PHILIPS 8" L	1	1	2
730-00256	BAG TOOL 12 X 10 X 5-1/4" FABRIC	1	1	2
730-00257	WRENCH 7/16" COMBO	2	2	4
730-00258	SKT DEEP 3/8" SQ DR 6-POINT 7/16"	2	2	4
730-00259	SKT DEEP 3/8" SQ DR 6-POINT 1"	2	2	4
730-00260	WRENCH 3/4" COMBO	2	2	4
730-00261	WRENCH 1/2" COMBO	2	2	4
730-00262	KIT HAND TOOLS 68 PIECE W/CASE	n/a	n/a	1
730-00263	SKT DEEP 1/2" SQ DR 6-POINT 1-1/8"	1	1	2
730-00264	SKT ADPTR 3/8" RATCHET TO 1/2" SKT	1	1	2
730-00266	WRENCH ADJ 1-5/16" MAX 10" L	2	2	4
730-00267	WRENCH 3/8" HEX L-KEY	1	1	2
730-00268	WRENCH 5/8" COMBO	2	2	4
730-00273	SKT RAIL 3/8" DRIVE HOLDS 9 SKTS	2	2	4
730-00291	Makita Drill/Driver Set	n/a	n/a	1
730-00292	Makita TW1000 impact driver	n/a	n/a	1
730-00303	Pick	n/a	n/a	1
730-00305	Shovel	n/a	n/a	2
730-00003	Sledge hammer	n/a	n/a	1
730-00169	UTILITY KNIFE	2	2	4
730-00170	WIRE STRIPERS	1	1	2
730-00307	WRENCH 3/16" HEX L-KEY	1	1	2
730-00306	WRENCH 5/16" HEX L-KEY	1	1	2

Packing List		
Consumables		Room
		Bag #3
Part #	Description	Quantity
200-00172	Lubricant	1
200-00160	Teflon tape	2
200-00131	Self-vulcanizing tape	2
200-00027	Zip ties	2
200-00104	Electrical tape	2
200-00049	Silicone rubber	1

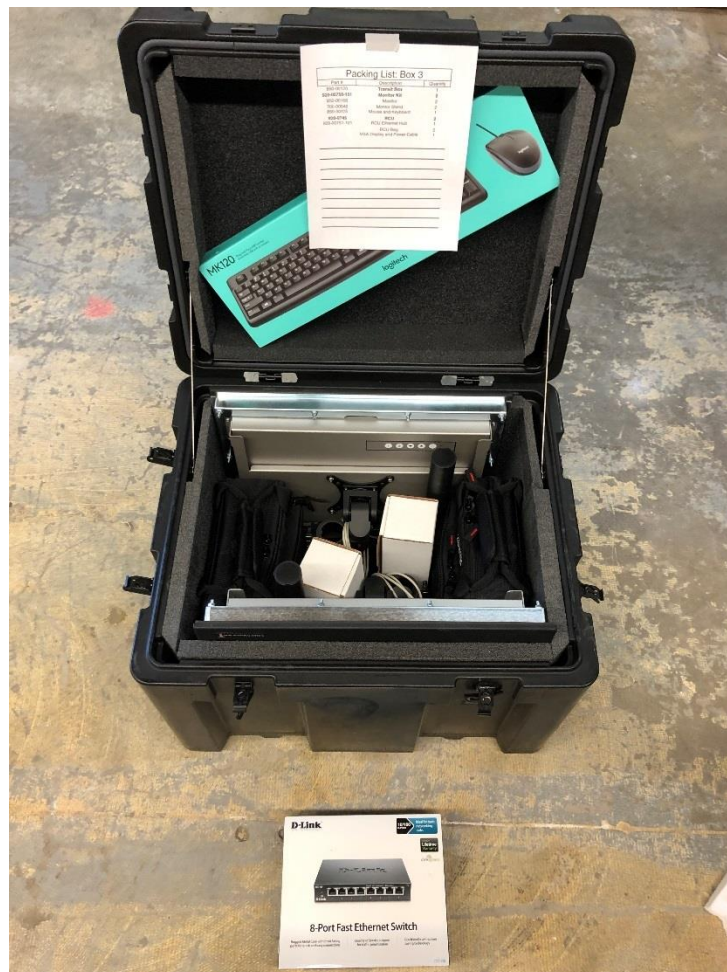
Packing List		
Box 1		LBS: 40
		KGS: 18
Part #	Description	Quantity
920-00256	Spare sensor	1



Packing List		
Box 2 / Monitors		LBS: 73
		KGS: 33
Part #	Description	Quantity
920-00750-101	Monitor Kit	2
850-00166	Monitor	2
700-00046	Monitor Stand	2
850-00125	Mouse and Keyboard	1
920-00745	RCU	2
	RCU Bag	2
	VGA Display and Power Cable	1



Packing List		
Box 3 / Monitors		LBS: 79
		KGS: 36
Part #	Description	Quantity
920-00750-101	Monitor Kit	2
850-00166	Monitor	2
700-00046	Monitor Stand	2
850-00125	Mouse and Keyboard	1
920-0745	RCU	2
920-00751-101	RCU Ethernet Hub	1
	RCU Bag	2
	VGA Display and Power Cable	1



Packing List		
Box 4		LBS: 49
		KGS: 22
Part #	Description	Quantity
920-00749-101	Entire Kit	1
280-00213-102	CA ASM N PLUG/ N PLUG RF	4
280-00198-103	CA ASM N PLUG TO N PLUG	3
	Ground to Air Radio Hardware	1
920-00744	Pair Wireless ACS	2
	Directional Antenna	2
	Wireless Radio Attachment Hardware	2
	Wireless Install Hardware	1
	Wireless Power Brick	2
	COM Radio Antenna Cable	1



Box 5		LBS: 75
		KGS: 34
Part #	Description	Quantity
750-00061	Theodolite Laser	1
	Safety Vest	2
	FM Transceiver	2
	Fiberglass Long Tape	1
	Tripod Lights	3
	FTM Cable	1
	1.5 V Alkaline Batteries	8



Packing List		
Box 6		LBS: 47
		KGS: 21
Part #	Description	Quantity
920-00563	Cal-Bit	1
920-00574-100	LNA	1
920-00787	Tilt Sensor Assembly	1
	Perpendicular Tilt Sensor	
	Parallel Tilt Sensor	
280-00571-103	OB-LT cable	2
240-00412	LNA and Cal/bit cable	2



Packing List		
Box 7		LBS: 68
		KGS: 45
Part #	Description	Quantity
190-00679	Sand Plate ASA/Tripod Leg	13
190-00680	Plates	3
890-00120	Transit Box	1
	Rope	1



Packing List		
Box 8		LBS: 65
		KGS: 29
Part #	Description	Quantity
920-00765-101	Tripod base	3



Packing List		
Box 9		LBS: 142
		KGS: 65
Part #	Description	Quantity
	Portable Nav Analyzer	1
920-00563	Spare Cal/Bit	1



Packing List		
Box 10		LBS: 110
		KGS: 50
Part #	Description	Quantity
730-00292	Makita TW1000 impact driver	1
730-00291	Makita drill/driver set	1
730-00262	Hand tools kit	1
190-00622	Rain hood	1
	Fluke clamp meter	1
	Tool kits	2
	Consumable kit	1



Packing List		
Box 11		LBS: 50
		KGS: 23
Part #	Description	Quantity
730-00002	Tension gauge	1
750-00028	ADIRpro prism	1
750-00027	Prism pole	1
790-00025	Rami ground to air antenna	1
	Ground to rod antenna rods	4
	Large tripod	1
	Mount rod	1
	Carabiner yellow rope	1
	Carabiner red rope	1



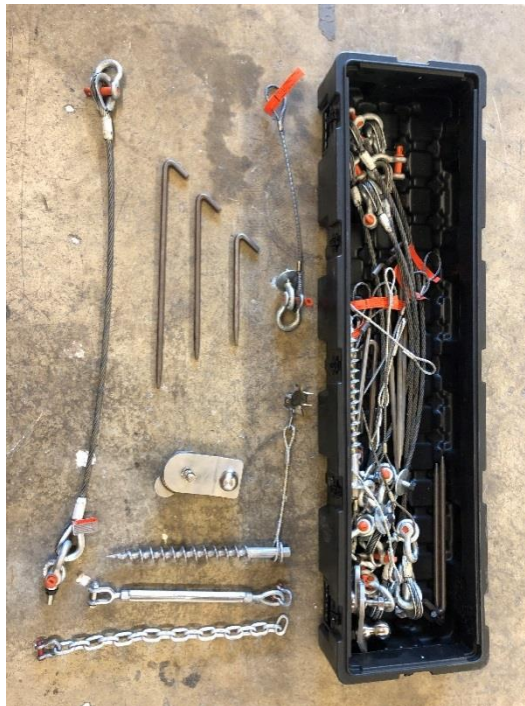
Packing List		
Box 12		LBS: 100
		KGS: 45
Part #	Description	Quantity
	ESA guywire bracket	1
	ASA turnbuckles	4
	Antenna structure turnbuckles	3
	Large augers	3
	Small augers	7
	Arrowhead anchors	6



Packing List		
Box 13		LBS: 140
		KGS: 64
Part #	Description	Quantity
920-00713	Winch assembly	1
	Mobile ground sticks kit	5
	Green poles	3
	Arrowhead hammer	1
	Air to Ground Antenna Mount	1



Packing List		
Box 14		LBS:
		KGS:
Part #	Description	Quantity
	Gin pole auger	2
090-00012	Turnbuckles	6
140-00119	Shackles	12
140-00120	2' chains	6
	Large shackle and cable ASSY	6
	Small shackle and cable ASSY	7
920-00788	Spar bracket	2
	Small stake	4
	Medium stake	4
	Large stake	4
190-00464	Small loops	3



APPENDIX B

Complete Container Packing Checklist**Container:**

- ☐ **Pre-Packing Procedure**
 - ☐ Antennas mounted
 - ☐ Cover screws tightened
 - ☐ Tripod brackets mounted
 - ☐ Rack cables organized
 - ☐ AC units strapped and remotes taped
 - ☐ AC units turned off
 - ☐ UPS powered off
 - ☐ All electronics powered off
- ☐ **Behind the Racks**
 - ☐ Boxes 5 and 8 loaded
 - ☐ Boxes 1 and 2 loaded
 - ☐ Boxes 3 and 6 loaded
 - ☐ Boxes 4 and 7 loaded
 - ☐ Boxes 1-8 padded and secured
- ☐ **The Tower Sections**
 - ☐ Tower sections loaded
 - ☐ Sledgehammer and pick loaded
 - ☐ Poles loaded in tower sections
 - ☐ Climbing harness packed
 - ☐ Guywires and cables secured to towers
 - ☐ Boxes 12 and 14 loaded under towers
 - ☐ ESA and ASA cabinets loaded and bolted down
 - ☐ Boxes 11 and 13 loaded on cabinets
 - ☐ Trolley loaded
- ☐ **The Doors**
 - ☐ Remaining cables reeled
 - ☐ Cables mounted to shelter doors
 - ☐ Shovels mounted to shelter doors
- ☐ **The Entrance**
 - ☐ Cal/BIT spool loaded and secured
 - ☐ Generator loaded and bolted down
 - ☐ Boxes 9 and 10 loaded onto generator
 - ☐ Fuel drum loaded and secured
 - ☐ Large sand coasters loaded and secured
 - ☐ Small sand coasters loaded and secured
 - ☐ Yellow ramps loaded and secured